

One-Pot Synthesis of Core–Shell Au@mSiO₂ Nanoparticles for Photothermal Applications

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Cite This: *ACS Appl. Nano Mater.* 2025, 8, 3631–3645



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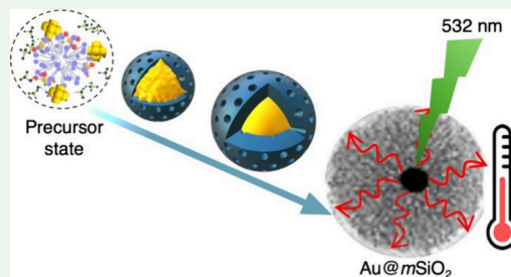
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ABSTRACT: Gold-mesoporous silica (Au@mSiO₂) core–shell nanoparticles (NPs) hold great promise for application in nanomedicine as they enable the photostimulated release of guest molecules located within their pores. In this study, we investigated the mechanism of a productive one-pot protocol for the preparation of core–shell Au@mSiO₂ NPs. The formation of gold NPs quickly followed by the formation of a silica shell eliminates the need to isolate and strongly protect the gold cores. A thorough mechanistic analysis provides insights into the relative kinetics of gold and silica formation, which can be fine-tuned to obtain Au@mSiO₂ smaller than 100 nm. The photothermal properties of such Au@mSiO₂ were characterized and modeled. Under the experimental conditions typically used for cargo release experiments, the model predicts a temperature increase for a single particle of less than 1×10^{-4} °C, demonstrating that the release is due to the collective heating of the medium by a myriad of NPs, rather than by the temperature elevation of individual NPs under irradiation. Overall, the studies developed here will enable better design and preparation of Au@mSiO₂ particles for diverse photothermal applications.

KEYWORDS: mesoporous silica nanoparticle, gold nanoparticle, core–shell nanoparticle, photothermal effect, mechanistic investigation, rational synthesis



INTRODUCTION

The design of nanocarriers or nanovehicles (NVs) for the transport of molecules (drugs, proteins, and mRNA) favors the carriage of molecules insoluble in biological media and provides an effective mode of localized delivery, minimizing secondary effects in treatments. Shipped in a nanovehicle, molecules are protected from degradation during transport and can be released upon reaching the target site. Liposomes, polymers, or solid nanoparticles have been explored as NVs. These include mesoporous oxides, which stand out for their low cost, good biocompatibility, and ease of surface modifications for targeted delivery.^{1,2}

For efficient transport, the NV must remain stable in the transport medium, preserving its cargo until reaching the point of delivery. Cargo release can be accomplished via particle erosion³ or by overcoming kinetic barriers through external stimuli such as chemical perturbations, magnetic fields, light, or ultrasound.^{4–6} Among these, light is an auspicious approach as it is noninvasive and may act in a confined space avoiding secondary effects.

Composite core–shell Au@mSiO₂ ions are promising NVs, combining the high loading capacity of mesoporous silica (mSiO₂) with the plasmonic properties of gold nanoparticles (AuNPs), which exhibit localized surface plasmon resonance (LSPR) in the visible spectrum. Upon irradiation at the LSPR wavelength, oscillating electrons on the particle surface transfer

energy to the metal lattice through electron–phonon coupling.⁷ This energy is then dissipated into the surrounding environment to raise the temperature, resulting in localized heating that operates in a different way than bulk heating.⁸

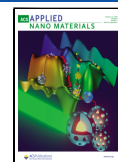
The vast knowledge for the bottom-up synthesis of mSiO₂ and AuNPs can be—in principle—taken to rationalize the synthesis of Au@mSiO₂ core–shell composites. On the one hand, for mSiO₂, the pioneering work on MCM-41⁹ paved the way for numerous strategies to produce mSiO₂ in solution at moderate temperatures using surfactant-based templates.^{10,11} The synthesis pathway involves hydrolysis and condensation of silicon alkoxides (usually TEOS) with further nucleation and growth of silica oligomers on the polar surface of micelles.^{12,13} Nowadays there are many accepted protocols that allow tuning the particle size, porosity, and porous structure, preserving the physicochemical stability and surface functionality of mSiO₂.^{2,14,15} Parameters such as the choice of amphiphilic template, pH, ethanol-to-water ratio, and TEOS-to-water ratio allow for fine-tuning both porosity and particle diameter.

Received: January 6, 2025

Revised: January 21, 2025

Accepted: January 23, 2025

Published: February 3, 2025



However, this is a kinetically controlled process and despite a great deal of work the mechanism has not yet been fully elucidated.^{10,11} On the other hand, gold NPs are biocompatible, photostable, exhibit low cytotoxicity,¹⁶ and have proven to be effective for photothermal conversion.^{17,18} Spherical AuNPs can be synthesized by robust bottom-up routes in solution, such as the widely used Turkevich–Frens method in aqueous solution.^{19,20} Briefly, a Au(III) salt or complex (i.e., HAuCl₄) is reduced by an appropriate reductant, resulting in rapid nucleation and growth of AuNPs. The reaction can be modulated by the speciation of Au(III) complexes which in turn depends on pH and ligand exchange.^{21,22} Even if the first stages of the reaction have been monitored by advanced characterization techniques,²³ the mechanism of NP growth and the influence of synthesis parameters on the final size and shape are still far from being fully understood.^{22,24}

Tailor-made design of Au@mSiO₂ for drug or protein delivery requires careful consideration of key parameters: (i) adjusting pore size to accommodate the cargo molecule; (ii) controlling the core–shell particle diameter so that it can effectively pass through the cell membrane, and (iii) optimizing the core's size and shape for efficient photothermal conversion. Au@mSiO₂ nanocomposites are typically synthesized through multistep processes that integrate AuNPs and mSiO₂²⁵ from the formation of gold seeds with well-defined size and shape. These particles are recovered, stabilized with appropriate capping, and placed in a reactor to which a template and silicon precursor are added, initiating hydrolysis and condensation to form silica oligomers that nucleate on the AuNPs, thus forming the core–shell structure.^{26–30} The method gives very good results, especially when the core is made of nanorods. However, the synthesis is conducted at high dilution to prevent AuNPs aggregation, which limits the possibility of scaling up production.

A more complex yet efficient approach is the one-pot synthesis of Au@mSiO₂ particles. In this method, the Au core is synthesized in situ in the same batch where the silicon precursor is further added.^{31–34} This one-pot procedure consumes less time and fewer reagents in order to fulfill the actual requirements of atom economy, yielding higher quantities compared to multistep processes, but the current literature (Table S1) shows limitations either because of a lack of reproducibility,^{35–37} the use of expensive precursors,³⁸ or a too high dilution^{33,34} which prevents efficient scaling up. Furthermore, there are no mechanistic studies that account for the interplay between AuNPs formation and mesoporous shell build-up.^{34,35,39}

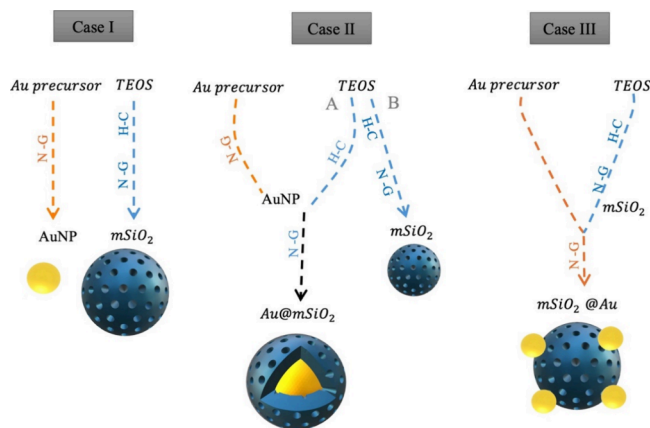
The aim of this work is to obtain NVs based on the synthesis of Au@mSiO₂ particles with spherical Au cores, particle diameter less than 100 nm so that they can pass through cell membranes with either small pores (2–3 nm) to enable molecular drug delivery or large ones (7–10 nm) able to accommodate and transport 40–70 kDa proteins and release the cargo under plasmonic heating. For this purpose, we have achieved a robust one-pot synthesis of Au@mSiO₂ particles suitable for photothermal conversion, using minimal reagents to approach sustainable chemistry principles.⁴⁰ Spherical Au cores were chosen as a model system that can provide interesting applications in photostimulated drug release for dermal treatment.^{41,42}

One challenge is to trigger both Au(III) reduction and TEOS hydrolysis and condensation simultaneously so that heterogeneous nucleation and growth of the mesoporous layer

occur on AuNPs. The key to this process is precise kinetic control, with an appropriate delay between AuNP formation and nucleation of silica oligomers. NaOH fulfills the conditions for an excellent trigger for the reaction in ethanol–water. Hydroxide ions (OH[−]) react with ethanol giving ethoxides, which act as a reductor for Au(III)⁴³ and promote TEOS hydrolysis and condensation.

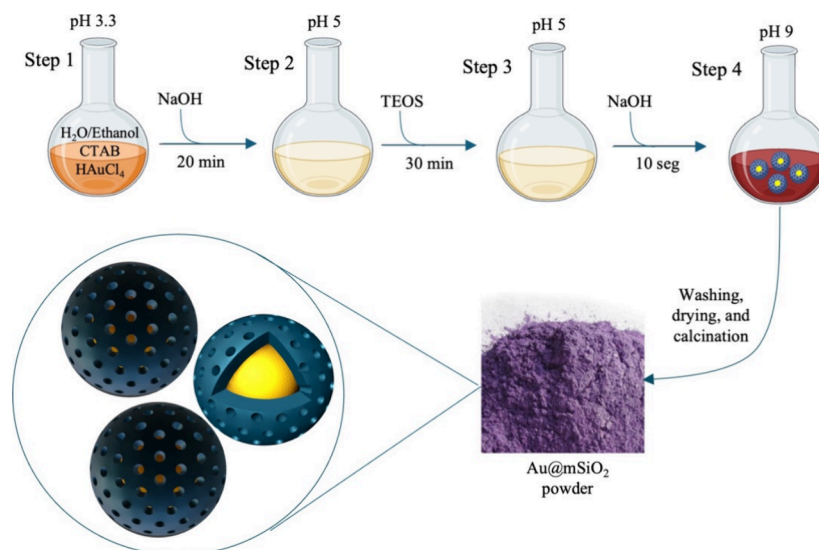
The minimal component synthesis reported in ref 31 produces a good yield of particles with diameters around 150–190 nm and CTAB molded pores of 2–3 nm that are not ideal for transporting cargo into cells. To solve this problem, we address the strategies usually employed in mSiO₂ synthesis to tune the particle and pore diameters. However, due to the presence of Au(III) and AuNPs, this methodology cannot be applied in a trivial way. In order not to fall into “trial and error”, our approach focuses on a thorough investigation of the mechanism of the synthesis with minimal components and then explores the changes of variables that lead to the desired product. Since in the proposed synthesis, there is a single trigger for the formation of AuNPs and mSiO₂, the formation of core–shell Au@mSiO₂ can be thought considering the coupling of two trajectories in the reaction space: (i) the transformation of Au(III) to AuNPs (gold trajectory) and (ii) the conversion of TEOS to mSiO₂ (silica trajectory). These trajectories initiate from a defined precursor state in the batch composed of Au(III), CTAB, and TEOS, and their intersection point in the reaction space defines the final product, as proposed in Scheme 1 for three paradigmatic cases.

Scheme 1. Possible Reaction Trajectories in the One-Pot Synthesis of Au@mSiO₂.^a



^aThe gold trajectory comprising nucleation and growth (N-G) is shown in orange, the silica trajectory comprising hydrolysis and condensation (H-C) and nucleation and growth (N-G) is shown in blue. *Case I*: gold and silica trajectories do not intersect. *Case II*: two scenarios are possible (A) gold trajectory develops before the intersection with silica trajectory leading to the expected core-shell structure; (B) concomitant homogeneous and heterogeneous nucleation of mSiO₂ leading to two NP populations. *Case III*: the intersection of trajectories occurs before nucleation and growth of gold leading to the inverse core-shell structure than *Case II*.

To achieve this goal, we performed a thorough analysis of the reaction mechanism by monitoring each stage with SEM, UV–vis spectroscopy, and pH on the one-pot synthesis of Au@mSiO₂ particles triggered by hydroxide ions (OH[−]) in alcohol–water. This analysis allows us to understand the interplay between the nucleation/growth of AuNPs and sol–

Scheme 2. Main Processes in One-Pot Synthesis of Au@mSiO₂ Core-Shell Particles with Minimum Components^a

^aThe overall process is made at 70 °C. A bright orange solution is obtained by adding HAuCl₄ to a 28% ethanol-water solution containing CTAB (Step 1). A first addition of NaOH (2.8 mM) leads to a pale-yellow solution (pH = 5) in Step 2. TEOS is added dropwise, with no apparent change (Step 3). In Step 4 a second addition of NaOH triggers the reaction (total Na⁺ concentration = 8.2 mM). After washing, drying, and calcination, a powder like the one shown in the picture is obtained.

gel chemistry for the mesoporous shell build-up, and then to find the appropriate parameters to control the particle sizes.

Another challenge involved is associated with cargo delivery by plasmonic heating, which requires a specific temperature at the surface of each particle. For correlating bulk and local temperatures, the macroscopic photothermal conversion of optimized NPs was measured and compared with the nanoscale modeling for single and multiple gold cores that may lead to hot spot enhancements. Overall, we aim to provide a rationalization of the mechanism in order to fine-tune the Au@mSiO₂ particle structure for specific applications and to understand the plasmonic heating process behind the photo-stimulated cargo delivery.³¹

EXPERIMENTAL SECTION

Hexadecyltrimethylammonium bromide (CTAB, A15235) was purchased from Alfa Aesar. Tetrachloroauric(III) acid trihydrate (HAuCl₄), tetraethyl orthosilicate (TEOS), hexadecyltrimethylammonium chloride 25 wt % in H₂O (CTAC), triethanolamine (TEOA), Brij C10, Pluronic F127, and all other reagents were purchased from Sigma-Aldrich. Gradient HPLC grade water (Fisher) and freshly prepared solutions of sodium hydroxide (2.0 M, NaOH) were used for all syntheses.

One-Pot Synthesis of Au@mSiO₂ Nanoparticles. In a 250 mL round-bottom flask, a mixture composed of CTAB (0.64 g, 1.8 mmol), water (100 mL), and ethanol (40 mL) was heated at 70 °C and stirred at 400 rpm using an olive stirring bar (12 × 25 mm). A tetrachloroauric solution (100 mM, 1.5 mL, 0.15 mmol) was added, and the mixture was stirred for 5 min at 600 rpm. A sodium hydroxide solution (2.0 M, 0.20 mL, 0.40 mmol) was then added, and the solution was stirred for 30 min. Then TEOS (1.0 mL, 4.5 mmol) was added dropwise over 2 min. Subsequently, a sodium hydroxide solution (2.0 M, 0.40 mL, 0.80 mmol) was added, which led to the discoloration, immediately followed by the appearance of blue to purple-colored species. The reaction mixture was stirred at 70 °C and 600 rpm for 2 h. The resulting Au@mSiO₂ were centrifuged at 5200g for 20 min, washed with ethanol twice, and dried at 60 °C. Finally, the template was removed by calcination at 500 °C for 2 h after a heating ramp of 10 °C/min.

To tune the characteristics of Au@mSiO₂, modifications were made to the previously described process. First, CTAB was combined with the cosurfactant Pluronic F127 (0.13 g in 1.69 mL water) or Brij C10 (0.13 g in 1.69 mL water for the 1:10 molar ratio or 0.25 g in 3.4 mL water for the 1:5 molar ratio). The cosurfactant solution was added to the reaction mixture 20 min after the first NaOH addition, and TEOS was added 15 min after the cosurfactant addition. Second, the synthesis with Brij C10:CTAB (1:5) was further studied by changing some parameters: change of alcohol, base source, and use of complexing agents. To study the effect of the alcohol, the ethanol was replaced by isopropanol or butanol (40 mL). To study the effect of the base source, the second amount of sodium hydroxide was replaced by a solution of sodium tetraborate (10 mL, 14 mmol) or by a solution of TEOA (0.20 mL of TEOA, 1.5 mmol). To study the effect of complexants, TEOA (0.20 mL, 1.5 mmol) was added 1 min after the second addition of sodium hydroxide. For all syntheses, the reaction mixture was kept for 2 h under stirring at 600 rpm. The resulting Au@mSiO₂ NPs were centrifuged, washed with ethanol, and calcined as described above.

Photothermal Experiments. A colloid solution (1.0 mg/mL) was formed by suspending 18.0 mg of Au@mSiO₂ synthesized with Brij C10:CTAB (1:5) in 18.0 g of water by sonication. These particles exhibit 2.5 ± 0.5 nm pores and an internal pore volume of 0.46 cm³/g. A quartz cuvette was filled with 2.0 mL of colloid, and a small magnetic stir bar was added. The mixture was kept under stirring while light (Spectra-Physics Millennia 532 nm CW, 0–800 mW) was shined from the top of the cuvette for 8 min, and then the laser was turned off. The temperature was monitored by using a digital thermometer with a type K probe.

Characterization. SEM images were recorded using a Zeiss Gemini ULTRA plus electron microscope operating at 10.0 kV and a working distance of 7 mm. A dilute nanoparticle suspension (3 μL) was dried on a doped silicon wafer for observation. TEM images were recorded using a Philips CM300 electron microscope operating at 300.0 kV. One drop of a dilute nanoparticle suspension was deposited on a carbon-coated copper grid and then dried for observation. Dynamic light scattering (DLS) measurements were conducted on a VASCO KinTM instrument at 25 °C. Zeta potentials (ζ) were determined using a Wallis analyzer (Cordouan, France) in 1 mM phosphate buffer pH 7, after checking that the NP size measured by DLS was less than 200 nm (DLS, cumulant fit). Absorption spectra were recorded by using a SAFAS Xenius XC Cuvette spectrofluor-

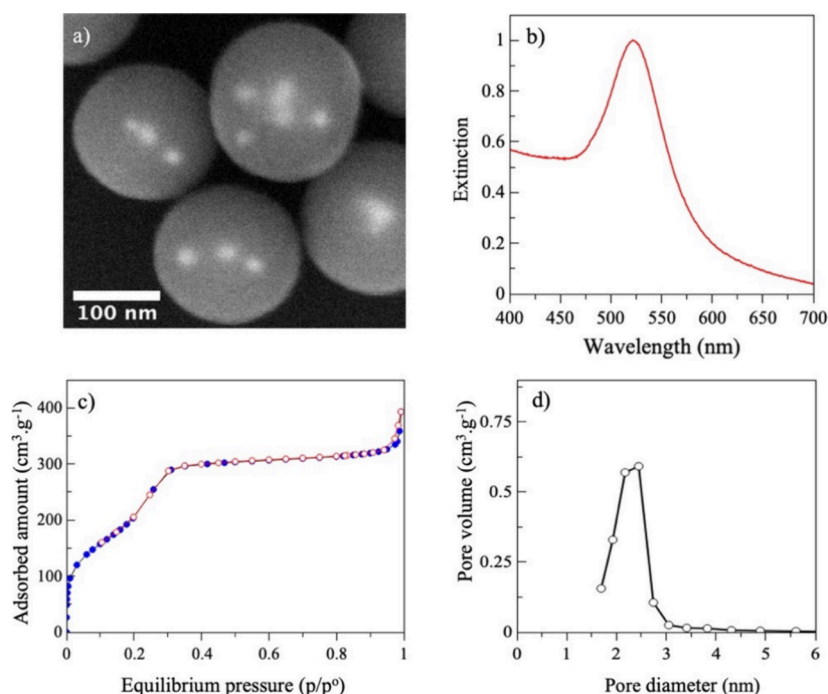


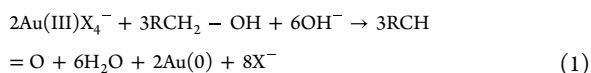
Figure 1. Characterization of Au@mSiO₂ obtained by the one-pot method. (a) SEM image, (b) UV-vis spectra, (c) N₂ adsorption (blue line)/desorption (red line) isotherms, and (d) pore size distribution.

ometer. FTIR spectra were measured using a Thermo Fisher Nicolet iN10 spectrometer in transmission mode on KBr pellets obtained with 0.5 mg of nanoparticles pressed within 140 mg of potassium bromide. Nitrogen sorption experiments were performed at 77 K using a Micromeritics Tristar 3000, after degassing samples 12 h at 80 °C under a secondary vacuum (10^{-5} Torr). The specific surface areas were calculated using the BET model in the range $0.15 < p/p^{\circ} < 0.35$ taking 0.162 nm^2 as the cross-sectional area for the adsorbed nitrogen molecules. The t -plot method was used to determine the micropore volumes, taking the Harkins and Jura equation as a reference sorption curve. The mesopore volumes were further determined by subtracting the obtained micropore volume from the full pore volume considered at $p/p^{\circ} = 0.7$.

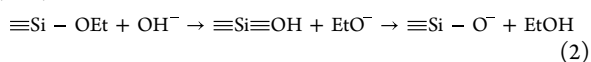
RESULTS AND DISCUSSION

One-Pot Synthesis of Au@mSiO₂ with Minimum Components. The reactants for the simplest way to prepare Au@mSiO₂ are CTAB, HAuCl₄, TEOS, and NaOH in 28% v/v ethanol:water. The process is carried out at 70 °C, which is optimal for *m*SiO₂ growth. This protocol, based on ref 31, is presented in Scheme 2. In the reaction medium, CTAB forms micelles as its concentration (12 mM) is higher than the first critical micelle concentration (cmc), as shown in Figure S1. Reactions 1–3 take place leading to the nucleation and growth of both Au and *m*SiO₂.

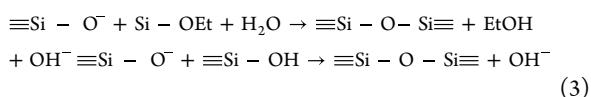
Gold reduction:



Hydrolysis:



Condensation:



Both components of the solvent mixture act as reactants. Reactions 1 and 2 are triggered by the increase of the hydroxyl concentration in the second NaOH addition (Step 4 in Scheme 2). Note that reaction 1 is in contradiction with the idea of “autoreduction” of Au(III) originally proposed.³¹ The success of core-shell particle formation depends on the mismatch between the reduction-nucleation-growth of AuNPs and the hydrolysis–condensation of TEOS (Case II in Scheme 1). Thus, the AuNPs form as the reaction is triggered in Step 4 (Scheme 2) and act as a nucleation center for the hydrolyzed TEOS units or oligomers. The molar fraction of ethanol in the solvent is 0.11 which, as pointed out by Quinson et al.,⁴⁴ is optimal for having enough reducing agents to allow reaction 1 to proceed and not having too much reducing agents to avoid overgrowth of the NPs. In addition, in the ethanol–water medium, the development of the mesoporous shell follows the path of the modified Stöber method leading to spherical MCM-41 particles.⁴⁵

In our hands, this synthesis gives on average 200 mg of Au@mSiO₂ particles with one or more $15 \pm 5 \text{ nm}$ gold cores at their center, for only 140 mL of solvent representing a 71% yield with respect to silica and 40% with respect to gold. This method thus enables an easy scalability compared to other methods reported so far (Table S1). The recovered Au@mSiO₂ particles, after drying and calcination at 500 °C, have an average diameter of $194 \pm 15 \text{ nm}$ as determined by SEM (Figure 1a) and exhibit the typical LSPR absorption at 525 nm in the UV-vis spectrum (Figure 1b). The core-shell particles exhibit a type IV N₂ sorption isotherm at 77 K (Figure 1c), from which we derived a specific surface area of $770 \text{ m}^2 \text{ g}^{-1}$ and a narrow mesoporosity centered at around 2.4 nm, consistent with other CTAB-templated mesoporous silica (Figure 1d).

Precursor State. We define as the precursor state the molecular configuration of the assembled reagents before triggering the reaction. The determination of the precursor

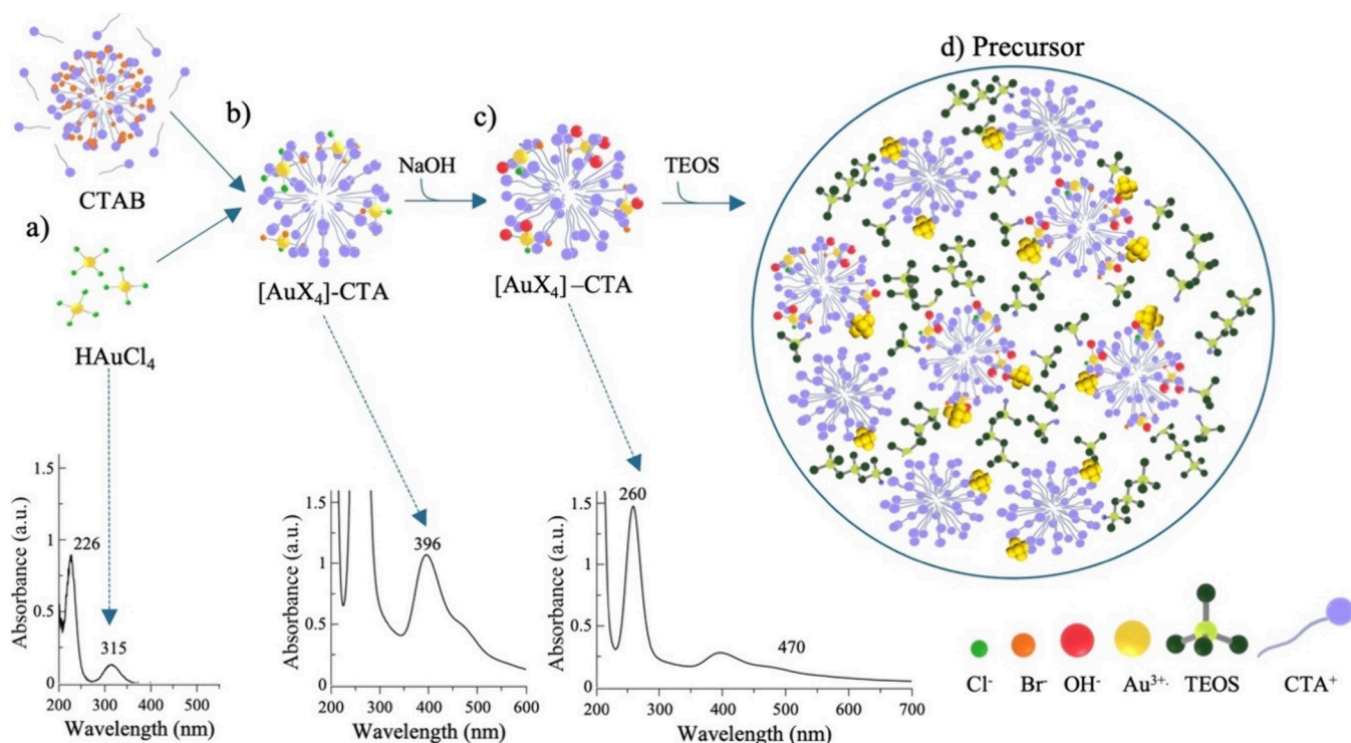


Figure 2. Scheme and UV–vis spectra at each step before precursor state formation. Initially, (a) HAuCl_4 in solution exhibits absorption bands at 226 and 315 nm. With the addition of CTAB in ethanol–water (b), the LMCT band shifts to 260 nm, and a new band appears at 396 nm, indicating substitution of Cl with Br and formation of $[\text{X}_4\text{Au}^- - \text{CTA}^+]$ complexes ($\text{X} = \text{Cl}$ or Br). Upon adding NaOH to adjust pH to 5 (c), the 260 nm band intensity decreases and the 396 nm band shifts to 470 nm, indicating Cl or Br substitution by OH in $[\text{X}_4\text{Au}^- - \text{CTA}^+]$ complexes and gold cluster formation. Addition of TEOS (d) shows no spectral change, indicating homogeneous dispersion of TEOS.

state was accomplished by monitoring the reaction from *Step 1* to *Step 3* in Scheme 2 by UV–vis spectroscopy complemented with SEM micrographs of aliquots freeze-dried at different stages of the reaction (Sections SH and SI in the SI).

Figure 2 schematizes the distribution of species prior to triggering the reaction that was determined by UV–vis spectroscopy. A 1 mM solution of HAuCl_4 in the reaction conditions (solvent and temperature) exhibits the LMCT band at 226 nm and a 315 nm weak band assigned to low probability $d-d$ transitions in symmetrical complexes (Figure 2a).⁴⁶ In the CTAB-containing solution, HAuCl_4 undergoes a fast partial ligand substitution of Cl by Br that shifts the LMCT from 226 to 260 nm and the 315 nm band to 396 nm with enhanced absorbance (Figure 2b,c).^{47,48} This enhancement of the absorption coefficient, resulting in an intense orange color, is attributed to the strong interaction between AuX_4^- and CTA^+ — denoted $[\text{X}_4\text{Au}^- - \text{CTA}^+]$ — which induces a geometric distortion favoring vibronic coupling of the Au–Br bond, assisted by the localization of AuX_4^- in the Stern layer of the micelle.^{21,49,50}

As the pH is increased to 5 after NaOH addition in *Step 2* of Scheme 2, the absorbance of the complex band at 396 nm decreases (Figure 2c), likely due to the substitution of Cl or Br ligands in $[\text{X}_4\text{Au}^- - \text{CTA}^+]$ by OH, resulting in a more rigid structure that relaxes the vibronic coupling. Additionally, the reduction of Au(III) to Au(I) or Au(0), probably induces the formation of Au clusters attached to the micelles.^{51,52} In *Step 3*, the addition of TEOS does not alter the UV–vis spectra denoting negligible TEOS – $[\text{X}_4\text{Au}^- - \text{CTA}^+]$ interactions. Indeed, the hydrolysis of TEOS at pH 5 would occur at a

very slow rate⁵³ thus, partially hydrolyzed TEOS available to interact with micelles can be disregarded.

In summary, the precursor state can be thought of as composed of a combination of $[\text{X}_4\text{Au}^- - \text{CTA}^+]$ complexes and a few gold clusters surrounded by micelles, with AuX_4^- distributed in solution (Figure 2d) (Au precursor in Scheme 1) and nonhydrolyzed TEOS. This establishes the initial condition for the reaction pathway.

From Precursor to Particle. Once the precursor state was established, we studied the progression of the reaction after triggering with NaOH to pH = 9.5 (*Step 4* of Scheme 2) by following the reaction with electron microscopy. The complete SEM follow-up is shown in Figure S12 and summarized in Figure 3. The process can be divided into three stages based on Au@mSiO₂ particle diameter (D_{part}) evolution, delimited by lines in Figure 3g, namely, *Stage I* with only AuNPs, *Stage II* where the mSiO₂ shell grows, and *Stage III* with constant particle diameter, as schematized in Figure 3h.

Stage I. The pH decay once the reaction is triggered in *Step 4* is considered an indirect parameter for comparing hydrolysis reaction rates.^{12,13} For the Au-free system in 28% (v/v) ethanol/water, the same initial slopes are obtained with and without CTAB (Figure S2), indicating that TEOS is homogeneously distributed within the solvent. This result differs from that found for the reaction in water, where TEOS is not soluble and is located inside the micelles.¹³ For the reaction route with Au(III), the pH decays more slowly than in the previous case (compare Figures S2 and S6), denoting a more complex pathway.

Nucleation and growth of AuNPs are the dominant features in this stage, while there is incipient hydrolysis and

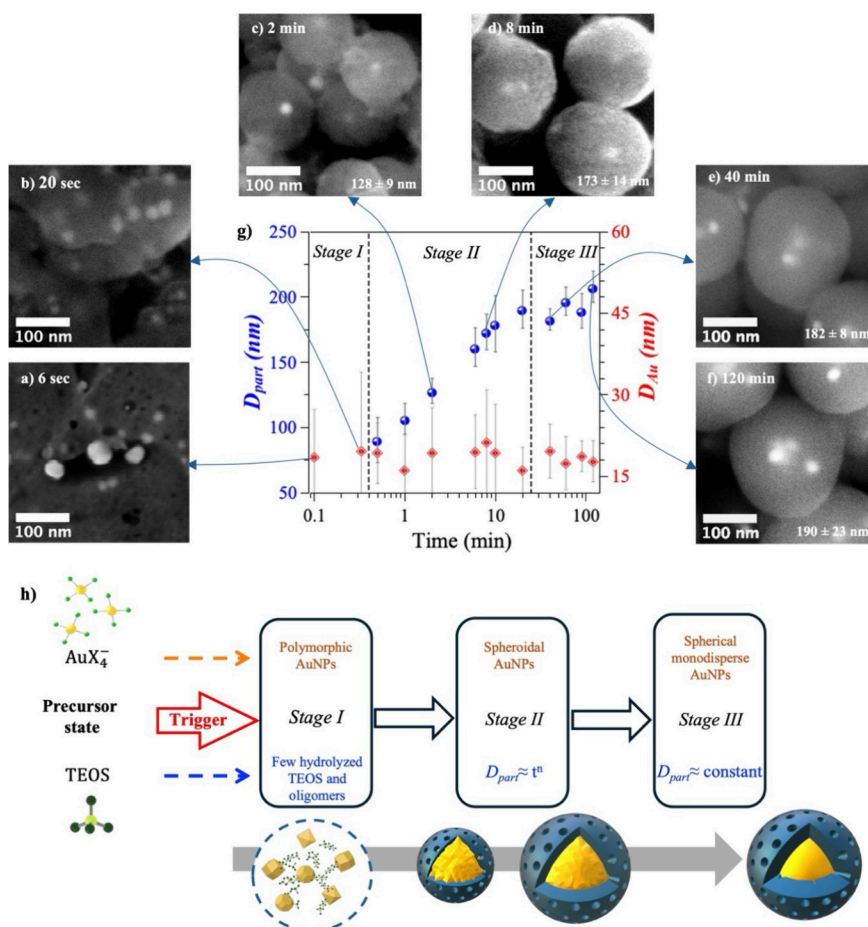


Figure 3. Representative SEM images of the reaction in the conditions presented in Scheme 2 (12:1:8.4 CTAB:HAuCl₄:NaOH in 28% ethanol–water). (a–f) Follow-up after triggering the reaction with NaOH. $t = 0$ corresponds to the moment at which NaOH is added in Step 4 (Scheme 2) increasing to pH = 9. (g) Evolution of D_{part} (blue, left axis) and D_{Au} (red, right axis). Dotted lines separate the three stages of the reaction according to the evolution of D_{part} (see text). (h) Main features of AuX₄⁻ and TEOS derived species at each stage of Au@mSiO₂ formation.

condensation of TEOS. The sudden change of pH produces the necessary concentration of reductant, EtO⁻, enabling the reduction of Au(III) species that will form nuclei. Several processes occur in less than 10 s as monitored by color changes from yellow to colorless, violet, and the final red-wine characteristic of spheroidal AuNPs (see the Supporting Video). Considering that AuNPs are formed as in Turkevich–Frens synthesis^{19,20,22} the ephemeral colorless batch is assigned to the presence of Au(I), and to a lesser extent, to the substitution of Cl or Br by OH in AuX₄⁻, which occurs at a slower rate than electron transfer reactions.^{48,22}

As seen in Figure 3a,b, AuNPs formed in this stage are polymorphic and polydisperse, consistent with the initial steps of the Turkevich–Frens synthesis.²² Once Au(0) is formed, heterogeneous nucleation proceeds more quickly on gold clusters at the micelle surface (Figure 2d). The AuNP growth is controlled by the low reactivity of OH-rich AuX₄⁻ at pH = 9.5 (X = Cl, Br, or OH). As the particle grows, it remains bound to CTA⁺ molecules in the micelles.²² The AuNPs surrounded by micelles and partially hydrolyzed TEOS are suitable seeds for the nucleation of mSiO₂ units.

From the absorbance at 400 nm, attributed to the 5d to 6ps interband transition of gold⁴⁴ (see Section SK in SI) we roughly estimate that in Stage I ca 5% of initial Au(III) was reduced to Au(0), leaving a high fraction of unreacted AuX₄⁻.

Stage II. This stage is dominated by the growth of the mesoporous shell up to a steady-state diameter with concomitant transformations on the AuNPs (Figure 3c,d,g). mSiO₂ nucleates on the AuNPs formed in Stage I, forming spherical particles. The time at which the particles start to be detected—ca 30 s after triggering (Figures S12c and S13b)—is close to the time at which the cloud point is observed in pH monitoring experiments under similar conditions without gold (Figure S3). pH drops from 9.7 to 8.4 over the course of the reaction at this stage (Figure S15). This pH variation is consistent with a silicate concentration at the end of Stage II of 7.7 mM, i.e., 24% of the initial TEOS under the hypothesis that only singly hydrolyzed species are present at Stage II.

In parallel, AuNPs evolve to spheroidal and more monodisperse NPs with a 15 nm average diameter. This process is also reported for the Turkevich–Frens synthesis.^{19–22,24} Also, the conversion of Au(III) to Au(0) goes from 5 to 50% as estimated from the absorbance at 400 nm (see Section SK in SI). Thus, as the NP diameter remains nearly constant, Au nucleation and growth also proceed in Stage II either in solution, at the surface of the growing mesoporous shell, or even inside the mSiO₂ shell (Figures S12 and S13). As proposed for the Turkevich–Frens synthesis, the formation of Au(I) by the comproportionation reaction²¹ should contribute to AuNP smoothing:



Reaction 4 provides Au(I) that may be further reduced on the AuNP. The participation of reaction 4 in the whole process allows to explain the slow pH decay in *Stages I* and *II*. As AuX_4^- is a square planar complex with a dimension lower than 0.5 nm,⁵⁴ unreacted AuX_4^- may diffuse through the silica shell — even when pores are clogged by CTAB — either from the outside solution or when trapped in the freshly formed silica shell.

For $m\text{SiO}_2$ particles, growth involves monomer addition, aggregation, coalescence, and dissolution that may occur simultaneously.⁵⁵ However, as pointed out by Semeykina and Zharov¹² the final particle size is mainly defined by the electrostatic or steric stabilization of the colloidal system and restricted by modifications of the surface with hydrophilic moieties or with surface complexes. In principle, the growth of the $m\text{SiO}_2$ shell should be independent of processes involving Au species. However, the complex pH drop when the reaction is triggered in the composite system (Figure S6) and the fact that the fraction of multiple gold cores increases at the end of *Stage II* (Figures 3 and S12) allow us to consider that Au species may exchange between particles.

Stage III. This stage is characterized by a nearly constant D_{part} which reflects a steady state between aggregation–condensation and dissolution processes. A relevant feature of this stage is the higher fraction of ovoidal particles with more than one AuNP as a consequence of the aggregation of the core–shell particles. At the same time, AuNPs embedded in $m\text{SiO}_2$ become more spheroidal and with lower size dispersion (Figure 3e–g), indicating that the dissolution of Au(0) by unreacted AuX_4^- takes place even at this stage. The ovoidal shape of the composite core–shell particles is also different from $m\text{SiO}_2$ synthesized in Au-free systems that are mainly spherical. At this stage, the Au@mSiO_2 particles reach sizes, surface charges, and structures for which colloidal stabilization is achieved.

Final Processing. Final processing involves washing, drying, and calcination for full removal of CTAB. The colloidal and hydrolytic stability of the final particles will allow their application to specific problems. After calcination, in addition to CTAB removal, the dehydroxylation of the silica surface is clearly detected in the FTIR spectra (Figure S17) by the decrease of the 963 cm^{-1} band. Importantly, no significant changes can be detected in D_{part} which means that the pore structure does not collapse under the applied thermal treatment.

Tuning the Particles through Synthesis Parameters.

Our interest is to control the size of Au@mSiO_2 particles in order to obtain monodisperse core–shell particles with a diameter (D_{part}) smaller than 100 nm that are suitable for crossing cell membranes and delivering cargo molecules inside plant or animal cells. For loading proteins or macromolecules, a second requirement is to achieve larger pores than those obtained with CTAB (Figure 1), i.e., $d_{\text{pore}} > 5\text{--}7\text{ nm}$. Finally, for a controlled plasmonic photothermal conversion, the necessary condition is to have monodisperse and single-shaped AuNPs. We thus first focus on spherical particles with LSPR absorption in the 520–530 nm range for photothermal conversion for cargo delivery useful in applications as dermal treatments.^{41,42}

As discussed in the previous section, for controlling the size of Au@mSiO_2 particles, it is necessary to restrict growth by

inhibiting aggregation and coalescence in *Stages II* and *III*. Based on the knowledge from mesoporous silica synthesis, to control particle size and to achieve pore enlargement, we explore (i) the use of larger templates than CTAB such as those nonionic from the Pluronics or Brij families or surfactant mixtures¹⁴ (ii) the use of bases able to trigger the reaction introducing steric hindrance or (iii) swelling the micelles, for example with alcohols.⁵⁶

Replacing CTAB with nonionic surfactants – Pluronic F127 and Brij C10 – results in polydisperse and nonreproducible particles. Better results were obtained with surfactant mixtures of CTAB–Pluronic F127 or CTAB–Brij C10 (Figure 4). The

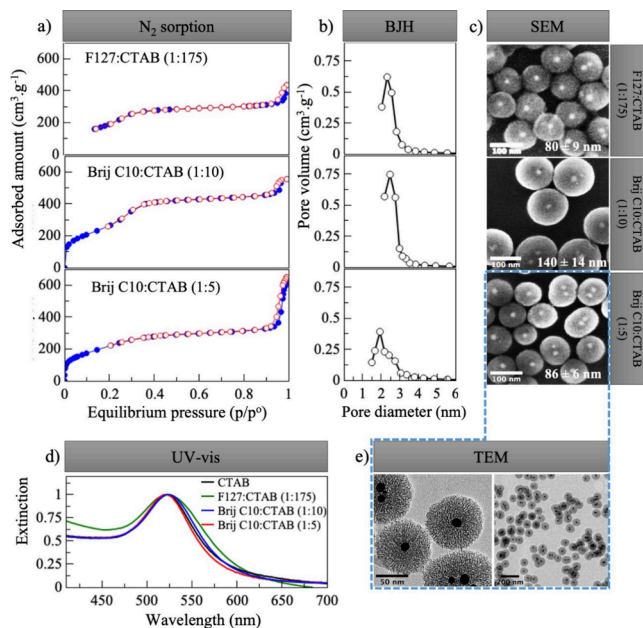


Figure 4. Characterization of Au@mSiO_2 obtained by the one-pot method using CTAB and cosurfactants. From the top to bottom: F127:CTAB (1:175), Brij C10:CTAB (1:10), and Brij C10:CTAB (1:5). (a) N_2 adsorption (blue line)/desorption (red line) isotherms. (b) Particle size distributions. (c) SEM micrographs, where the white dots correspond to the gold cores and the gray zones correspond to the silica shells. (d) UV–vis spectra of Au@mSiO_2 obtained using CTAB and surfactant–CTAB mixtures. (e) TEM micrographs of Au@mSiO_2 synthesized with BrijC10:CTAB (1:5) at different magnifications. The black dots correspond to the gold core and the gray area to the silica shells.

incorporation of Pluronic F127 or Brij C10, as cosurfactants, in *Step 2* (Scheme 2) indeed results in a decrease in shell thickness, as seen in the SEM micrographs in Figure 4 for F127:CTAB, Brij C10:CTAB. The Brij C10:CTAB molar ratio also has an influence on D_{part} , with size increasing from $86 \pm 6\text{ nm}$ to $140 \pm 14\text{ nm}$ when the amount of Brij C10 is halved (Figure 4c). Figure 4a,b presents the N_2 sorption isotherms and pore size distributions derived from the BJH analysis. The textural properties of the Au@mSiO_2 particles synthesized with different cosurfactants are compiled in Table 1. Regardless of the cosurfactant used with CTAB, the obtained particles exhibit homogeneous and cylindrical mesopores around 2–3 nm. Detailed data analysis of N_2 adsorption isotherms can be found in the Section SC in the SI. The extinction spectra of the core–shell particles obtained with the different surfactant–CTAB mixtures exhibit a

Table 1. Textural Parameters Derived from the N₂-Sorption Isotherms for Samples Synthesized with Different Surfactant-CTAB Mixtures^a

surfactant	S_{BET} (m ² /g)	mesopore volume (cm ³ /g)	BJH pore diameter (nm)	Gurwich pore diameter (nm)
CTAB	757	0.47	2.3	2.5
F127:CTAB (1:175)	778	0.45	2.3	2.3
Brij C10:CTAB (1:10)	988	0.66	2.5	2.7
Brij C10:CTAB (1:5)	790	0.46	2.0	2.3
Brij C10:CTAB:TEOA (1:5:4)	988	0.64	2.0–3.0	2.6

^aThe pore volume and Gurwich average pore diameters (4 V/s) were calculated assuming cylindrical pores, considering the adsorbed amount at $p/p^{\circ} = 0.6$, after capillary condensation completion (see the Section SC in the SI).

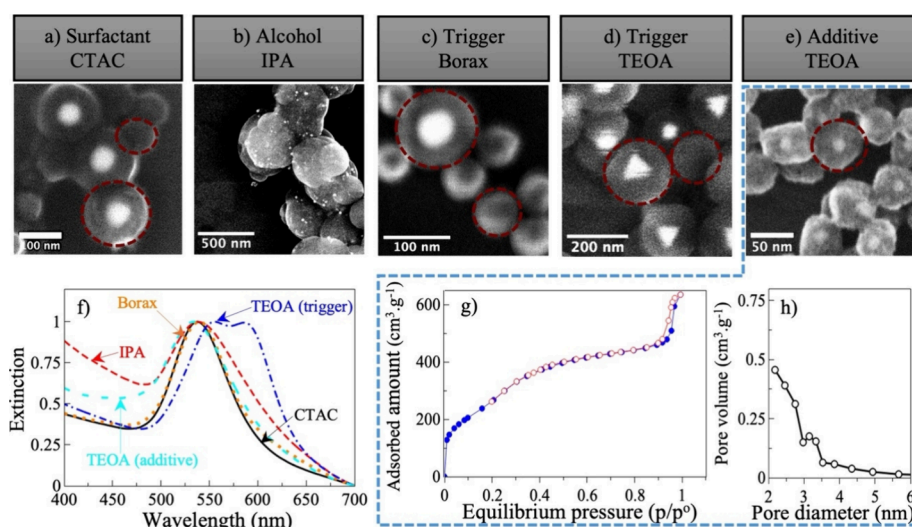


Figure 5. Characterization of Au@mSiO₂ with different synthesis parameters. SEM micrographs of Au@mSiO₂ synthesized with (a) Brij C10:CTAC (1:5) instead of CTAB. (b) IPA instead of ethanol. (c) sodium tetraborate instead of NaOH. (d) TEOA instead of NaOH. (e) TEOA as additive added 1 min after triggering the reaction with NaOH. (f) Normalized UV–vis spectra of calcined Au@mSiO₂. (g) N₂ adsorption (red line)/desorption (blue line) isotherms and (h) BJH pore size distribution of calcined Au@mSiO₂ synthesized using NaOH as reaction trigger and TEOA as additive.

maximum at 525 nm, as expected for 15 nm diameter spherical AuNPs coated by *m*SiO₂ (Figure 4d).

Both Pluronic F127 and Brij C10 form hybrid micelles with CTAB that modify the shell template and may act as surface modifiers, reducing the interaction between silicate species and the growing mesoporous shell, which may limit its growth.¹⁴ Although hybrid micelle formation could alter the synthesis mechanism, monitoring the Brij C10 (1:5) reaction via SEM, UV–vis, and pH showed no significant effect on precursor formation or AuNPs synthesis.

The pH decay in Brij C10:CTAB (1:5) closely matches that of CTAB alone (compare Figures S2 and S3), indicating that hybrid micelles do not affect the initial TEOS hydrolysis and condensation steps. The cloud point observed during pH monitoring aligns with the detection of initial particles by SEM (Figures S3 and S13). The UV–vis spectrum of the precursor state, with or without Brij C10, remains nearly identical (Figure S11), confirming that Brij C10 does not impact the precursor state.

Based on the results obtained for the cosurfactant:CTAB compositions in Figure 4 and compiled in Table S2, we consider the synthesis with Brij C10:CTAB (1:5) whose TEM images are shown in Figure 4e, is the optimal one in this series as it fulfills the set requirements: it is statistically reproducible, provides calcined Au@mSiO₂ particles with $D_{\text{part}} < 100$ nm, and exhibit good colloidal stability in aqueous media. In what

follows, we analyze the role of other synthesis variables for this optimized synthesis.

The role of the surfactant in the precursor state can be better understood by replacing CTAB with CTAC in the 1:5 molar ratio mixture with Brij C10. Two kinds of particles are obtained: those without an Au core ($D_{\text{part}} \sim 87$ nm) and those with an Au core ($D_{\text{part}} \sim 150$ nm), as seen in Figure 5a. CTAC and CTAB differ in the counterion Cl[−] being a harder anion than Br[−], thus affecting nucleation of AuNPs with less nuclei what results in larger Au cores coated by *m*SiO₂ that are formed concomitantly with pure *m*SiO₂ NPs, as presented in Case II of Scheme 1.

Alcohols have an influence on the AuNP nucleation and growth,⁴⁴ as well as in the development of *m*SiO₂.^{56,57} We explored the role of increasing the alcohol chain length from C₂ to C₄ in the final calcined particles. With butanol, no purple color was detected in the final powder, and large *m*SiO₂ particles were formed, as observed by DLS (Table S2). With isopropyl alcohol (IPA), large (150–500 nm) yet polydisperse *m*SiO₂ particles were obtained, which featured polydisperse AuNPs (10–20 nm) at their surface (Figure 5b). Comparing the reaction in ethanol and IPA, the emergence of the typical red color of AuNPs is much slower in IPA than in ethanol (see Figure S7), but the hydrolysis of TEOS monitored by pH decrease occurs at similar rates in ethanol and IPA (Figure S4). In this case, the crossing of the trajectories in the reaction space occurs after the formation of mesoporous particles, on

which there is a heterogeneous nucleation of the Au precursors, as shown in *Case III* of Scheme 1.

The base used to trigger the reaction (*Step 4*, Scheme 2) defines the pH and the ionic strength and may introduce steric factors controlling the interactions between nuclei and particles. pH affects the kinetics of $m\text{SiO}_2$ hydrolysis and condensation as well as Au(III) speciation and reduction potential. As the optimal pH for the synthesis of mesoporous silica particles falls in the range of 8.5–10.5,⁵⁸ we analyzed the effects of sodium tetraborate (Borax, $\text{pK}_a \sim 9.2$) and triethanolamine (TEOA, $\text{pK}_a \sim 7.8$) while maintaining the pH of the reaction nearly constant. Triggering the reaction with borax at pH 9.0 led to a bimodal distribution with Au@ $m\text{SiO}_2$ particles having $D_{\text{part}} = 120 \pm 9$ nm and $m\text{SiO}_2$ particles without gold core ($D_{\text{part}} = 62 \pm 9$ nm) (Figure 5c). Borax alters the nucleation and growth of AuNPs and develops large gold cores ($D_{\text{Au}} = 35 \pm 6$ nm). Here fewer gold nuclei were formed, and growth leads to large particles, probably because borax introduces steric hindrance for gold nucleation. In this scenario, the nucleation of silicate oligomers occurs both homogeneously and heterogeneously on AuNPs, giving the bimodal distribution of particles, consistent with trajectories that cross in the reaction space as in *Case II* (Scheme 1).

TEOA is a weak base ($\text{pK}_a \sim 7.8$) that regulates silica hydrolysis and condensation by the formation of atrane species, allowing control of particle size and often leading to well-dispersible nano-objects.⁵⁸ Triggering the reaction with TEOA instead of NaOH (Figure 5d) results in $m\text{SiO}_2$ particles with $D_{\text{part}} = 60 \pm 8$ nm without a gold core, and predominantly triangular gold nanoplates with a $m\text{SiO}_2$ shell and $D_{\text{part}} = 106 \pm 9$ nm. The triangular gold shape introduces anisotropy that gives two LSPR bands at 550 and 590 nm (Figure 5f). The development of triangular nanoplates indicates that TEOA participates in *Stage I* as a shape-directing agent. This aligns well with previous reports where triangular gold nanoplates were obtained by using CTAB/TEOA mixtures.⁵⁹ The lack of enough AuNP seeds means that $m\text{SiO}_2$ forms through both homogeneous and heterogeneous nucleation consistent with *Case II* (Scheme 1).

Taking advantage of the possible formation of surface silatranes with TEOA,⁶⁰ the synthesis with Brij C10:CTAB (1:5) was performed in the usual synthetic pathway adding TEOA (10.3 mM) in the progress of *Stage II*. The addition of TEOA 1 min after triggering the reaction with NaOH led to monodisperse Au@ $m\text{SiO}_2$ NPs with $D_{\text{part}} = 50 \pm 8$ nm while preserving the 15 nm gold cores (Figure 5e) and maintaining the characteristic LSPR band (Figure 5f). The broader pore size distribution between 2.0 and 3.5 nm derived from N_2 -sorption isotherms (Figure 5g,h and Table 1) should be a consequence of silica dissolution from the pore walls, as reported for mesoporous films in weak basic media.⁶¹ The hydrolysis of TEOS remains unaltered, as the same slope for pH decrease was found (Figure S5). The time between NaOH triggering and TEOA addition plays an important role, as TEOA inhibits growth from the diameter reached at the time of its addition. In fact, D_{part} remained unaltered when TEOA was added 10 min after the triggering with NaOH (Figure S14).

Other synthesis parameters, such as reaction temperature and gold precursor concentration, were analyzed (see Section SG in the SI). The temperature was systematically varied within the range of 50 to 80 °C, resulting in negligible variations in the characteristics of the calcined Au@ $m\text{SiO}_2$

particles (Figure S8). Regarding the concentration of the gold precursor, there were no differences in D_{part} nor D_{Au} ; however, two types of particles were observed when the precursor concentration was halved, yielding both Au@ $m\text{SiO}_2$ and empty $m\text{SiO}_2$ particles (Figure S9).

The narrow margin to modify the synthesis variables without altering the precursor state or *Stage I*, as in the case of IPA mainly affects the Au precursor trajectory. In contrast, when the synthesis variables act on *Stage II*, as in the case of hybrid micelles, monodisperse core-shell particles with $D_{\text{part}} < 100$ nm are obtained. Based on this evidence, we can infer that the optimal crossing of trajectories is found on the time scale between *Stage I* and *Stage II*. However, none of these synthesis variables allow us to obtain Au@ $m\text{SiO}_2$ particles with larger pores, which we successfully obtained through a two-step procedure from the tiny Au@ $m\text{SiO}_2$ NPs prepared with TEOA.⁶² These results will be presented in a subsequent report.

Photothermal Conversion. The photothermal heating of Au@ $m\text{SiO}_2$ was assessed using two distinct approaches. The first approach aimed to characterize plasmonic heating at the macroscale by experimentally measuring the temperature increase upon the irradiation of the particles with a 532 nm laser. This result is counteracted with the second approach where theoretical modeling is used to examine heating at the nanoscale, with particular emphasis on how factors such as particle size, porosity, and the number of gold cores per particle influence the light-induced heating of Au@ $m\text{SiO}_2$.

Heating at the Macroscale. The ability of the Au@ $m\text{SiO}_2$ NPs to dissipate heat was probed using Au@ $m\text{SiO}_2$ with $D_{\text{part}} = 86 \pm 6$ nm and $D_{\text{Au}} = 15 \pm 5$ nm. These particles correspond to the synthesis using CTAB:Brij (1:5) (Figure 4e and Table 1) since they were the only type of particles that fulfilled one of the aims of this work ($D_{\text{part}} < 100$ nm). The experiment involved irradiating a stirred colloid with a collimated laser beam (2.3 mm in diameter, 532 nm wavelength). The temperature increase was monitored over time during both the heating and cooling phases (Figure 6a). Upon irradiation, the temperature followed an increase that could be fitted by an exponential function. After cessation of irradiation at $t_1 = 8$ min, the temperature decreased according to a monoexponential decay as described previously.^{63,64} The maximum temperature increase after 8 min of irradiation was proportional to the

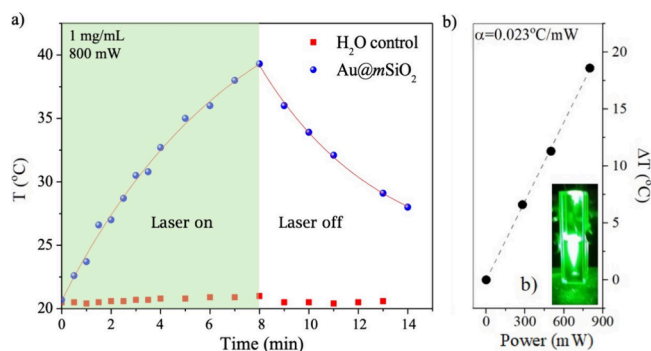


Figure 6. (a) Temperature variation of Au@ $m\text{SiO}_2$ colloid at 1.0 mg/mL (9×10^{11} NP/mL) under a collimated 532 nm laser irradiation at 800 mW for 8 min (blue dots) with water as a control (red squares). (b) Temperature increase as a function of laser power. Inset: the picture shows the scattered green light during the heating experiment with the exciting beam coming from the top of the cuvette.

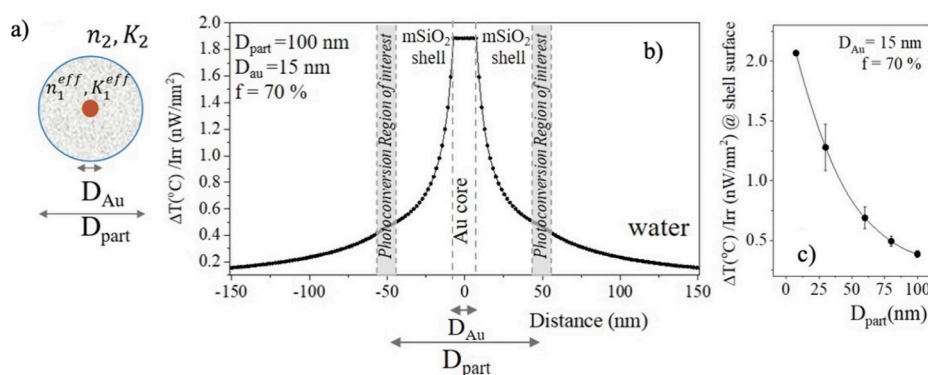


Figure 7. (a) Scheme of the Au@mSiO₂ particles and parameters used in simulations, (b) ΔT profile per irradiance ($\frac{nW}{nm^2}$) for an Au@mSiO₂ with $D_{Au} = 15$ nm, $D_{part} = 100$ nm, and $f = 70\%$, (c) ΔT at the surface as a function of D_{part} , with $D_{Au} = 15$ nm, and $f = 70\%$, error bars correspond to the variation inside the photoconversion region of interest.

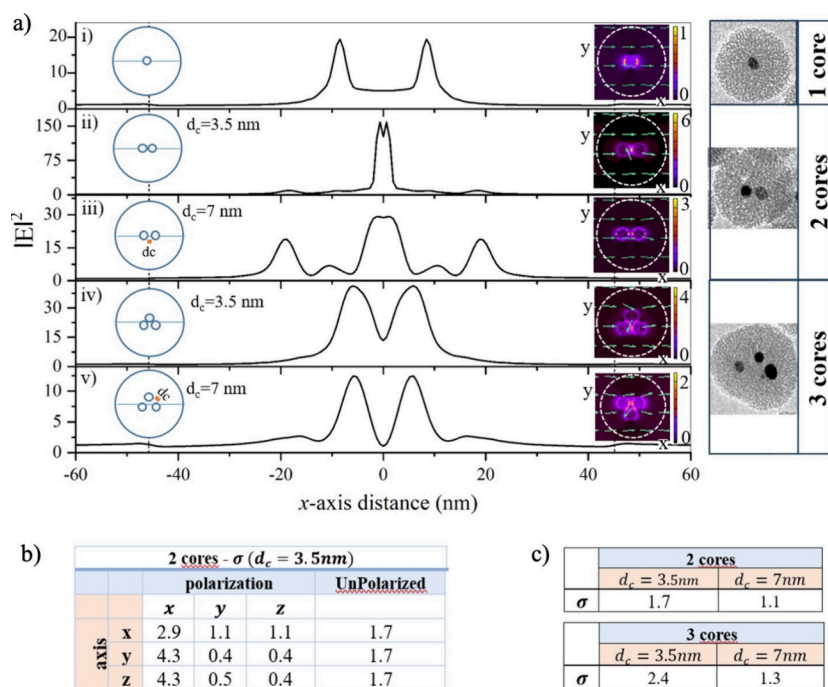


Figure 8. (a) $|E|^2$ profiles along the x -axis under x -polarization for Au@mSiO₂ particles with (i) 1 core, (ii) 2 cores with an intercore distance $d_c = 3.5$ nm, (iii) 2 cores with $d_c = 7$ nm, (iv) 3 cores with $d_c = 3.5$ nm, and (v) 3 cores with $d_c = 7$ nm. For each configuration, the left inset displays the corresponding schematic, the right inset shows a TEM image, and the middle panel provides the 2D x - y $|E|^2$ distribution, normalized to the maximum of the $|E|_{INP}^2$ case for x polarization. (b) Calculated σ values from the simulated $|E|^2$ data in (a-i) discriminated for each polarization direction calculated along the 3 main axes. The last column represents the average for unpolarized light. (c) Average σ values in polarization states and directions for all cases in (a), considering $d_c = 3.5$ nm and $d_c = 7$ nm.

incident power (Figure 6b), in accordance with what has been described in the literature.⁶³ The maximum temperature is nearly independent of the concentration in the 0.25–1.0 mg/mL range (Figure S18), which can be understood as most of the incident light is absorbed by the sample through the 20 mm of the solution, as seen in the cuvette photograph (Figure 6b). Moreover, the cuvette used and the water itself could not absorb light, leading to a negligible temperature increase when no Au@mSiO₂ was used. These experiments confirm the potential of Au@mSiO₂ as strong absorbers for photothermal therapy or photothermal release of guest molecules included in the mesopores.

Following the heat transfer balance described (See Section SN in SI), we found photothermal efficiency conversion values around 0.7 for Au@mSiO₂ (Table S5), which is consistent with

values reported for similar core–shell nanoparticles.⁶⁴ Bare AuNPs exhibited slightly higher efficiencies, ranging from 0.7 to 0.8 (Table S5). The fact that the efficiency is below 1 underscores the contribution of particle scattering, which leads to light losses to the top and sides of the cuvette, as is evident from the experimental setup (Figure 6b).

Modeling of the Plasmonic Heating at the Nanoscale. The physical basis for photothermal conversion for bare AuNPs is well-known.⁶⁵ To calculate the photothermal response of core–shell particles we have adapted previous results⁶⁶ to consider a spherical Au metal core of diameter D_{Au} and thermal conductivity K_0 , surrounded by a porous mSiO₂ shell of diameter D_{part} and thermal conductivity K_1^{eff} . As the shell is mesoporous, both the refractive index and effective conductivity of the mSiO₂ part have been calculated by

applying EMA models^{67,68} considering the porosity of the shell. Pores were considered to host water as well as the surrounding environment. In all cases, $K_0 > K_1^{\text{eff}} \geq K_2$ where K_2 is the thermal conductivity of the external aqueous medium. This system is schematized in Figure 7a. Figure 7b exhibits the photothermal response as a function of distance for the system used in the photothermal experiment ($D_{\text{Au}} = 15$ nm, $f = 70\%$, and $D_{\text{part}} = 100$ nm). At the surface of the particle ($d = D_{\text{part}}$) for an irradiance of $I_{\text{rr}} = 1$ nW nm⁻², we calculated $\Delta T \approx 0.35$ °C. It is worth noticing that the irradiance used for the photothermal experiments was below 0.2 W mm⁻², i.e., 2×10^{-4} nW nm⁻². Under this irradiance, the expected temperature increase for $D_{\text{part}} = 100$ nm is $\Delta T \approx 7 \times 10^{-5}$ °C under isothermal conditions. Therefore, the observed increase in temperature can only be explained by a collective heating of the medium by numerous NPs.

As seen in Table S2, the use of diverse surfactants, alcohols, and bases may affect the morphology of the particles affecting D_{Au} or D_{part} . To consider this diversity of sizes, the effect of the shell thickness was evaluated (Figure 7c). For the smallest $D_{\text{part}} = (50 \pm 8)$ nm, the temperature at the particle surface could rise up to $\Delta T \approx 0.8$ °C nm² nW⁻¹. To complete the understanding of the role of each morphological parameter in the temperature, variations in porosity have been simulated (Figure S19b) yielding a maximum difference of $\Delta T \approx 0.15$ °C nm² nW⁻¹ between dense SiO₂ shell ($f = 0\%$) and the absence of it ($f = 100\%$) at the surface of particles of $D_{\text{part}} = 100$ nm. More important differences could be observed for larger cores which could raise up to $\Delta T \approx 5$ °C nm² nW⁻¹ for particles of $D_{\text{Au}} = 30$ nm (Figure S19a). However, all the individual ΔT variations would not allow a sufficient temperature increase to enable a photothermally stimulated release of guests at the irradiance employed under the usual experimental conditions (<1 W mm⁻²). Nonetheless, each tiny NP surface temperature variation contributes to the overall increase of temperature in the system, since it is possible to optically stimulate many NPs at the same time.

From TEM images of the core-shell particles (Figure 8a, thick insets), some particles exhibit 2 or more cores. From microscopy statistical analysis on the sample in Figure 4e, 61% of the NPs possess a single core, while 39% have multiple cores. To understand the role of multicore NPs in the photothermal response, simulations have been performed using *Photonic open source code*.⁶⁹ Particles with 2 and 3 cores have been considered since they are statistically more frequent in the samples, with a mean distance between cores of $d_c = 7 \pm 3$ nm, as measured in TEM micrographs.

Heat power density comes from the Joule effect, $q(r) = \frac{1}{2} \text{Re}[J^*(r) \cdot E(r)]$, with $J(r)$ the current density and $E(r)$ the electric field. For harmonic waves, this expression results proportional to the electric field density $|E|^2$. For each of the configurations, we have calculated the $|E|^2$ distribution in the 3D space for polarization in the x , y , and z directions since the morphology is no longer isotropic. To show the behavior of $|E|^2$, we have selected an x -axis profile for x polarization, for 1, 2, and 3 cores at two different intercore distances (d_c). Figure 8a-i shows the dipolar distribution of $|E|^2$ for 1 core, where the EM field is concentrated in the neighborhood of the metal surface. For particles with 2 cores, the $|E|^2$ distribution is very sensitive to the intercore distances. Figure 8a-ii,iii show the $|E|^2$ profile in the x -axis for $d_c = 3.5$ nm and $d_c = 7$ nm, respectively. In both cases, the 2D x - y distribution is included

showing a clear hot spot for the shortest distance, with high energy concentration at the center of the particles of almost 1 order of magnitude with respect to $|E|^2$ in Figure 8a-i. Particles composed of 3 cores show the same behavior for $d_c = 3.5$ nm (Figure 8a-iv,v), where $|E|^2$ is more localized and intense than for $d_c = 7$ nm (Figure 8a-v), which can also be observed in the corresponding 2D x - y $|E|^2$ distributions. $|E|^2$ has also been performed for the other 2 polarization cases, with specific parameters of simulations displayed in Section SN in SI. To calculate the temperature at the surface of the Au@mSiO₂ particle it is necessary to take into account all the $q(r)$ contributions to the $\Delta T(r = \frac{D_{\text{part}}}{2})$. To take into consideration all contributions, temperature variations can be calculated as $\Delta T(\frac{D_{\text{part}}}{2}) = \int G(r, \frac{D_{\text{part}}}{2}) q(r) dr$, where $G(r, \frac{D}{2})$ represents the Green function associated with the Poisson equation.

To understand if the number of cores modifies the temperature at $r = \frac{d_{D_{\text{part}}}}{2}$, the ratio $\sigma = \frac{\Delta T}{\Delta T_1}$ has been calculated for each of the configurations of Figure 8a, where ΔT_1 represents the variation of temperature for only 1 core. As a first approach, σ has been calculated using a 1D Green function (see Section SN in SI) at axes x , y , and z as privileged directions of the system, due to the configurations of cores at the center. As the $|E|^2$ distribution is higher at those axes, it is expected to be upper bounds for the temperature. Figure 8b shows the values obtained for σ in particular for the 2 cores with $d_c = 3.5$ nm case for the 3 axes in the 3 polarization cases. Due to the distribution of the $|E|^2$ density in x -polarization (which can be observed in the corresponding 2D graph), it is possible to observe a clear increment of the temperature with respect to the 1 core case, while in y and z polarizations, this effect does not occur and in fact, it represents less temperature for some axes contributions. Although an expected anisotropic response is found, photothermal experiments have been done with unpolarized light and with randomly distributed particles, which implies averaging the polarization states and directions. These averages (Figure 8c) imply a temperature increment with $\sigma = 1.7$ in particular for the 2 cores with $d_c = 3.5$ nm. For other multicore particles studied where hot spots are present ($d_c = 3.5$ nm), ΔT is higher than for cores that are separated by 7 nm for which there is almost no increment in ΔT since σ is close to 1.

Simulations thus show that it is possible to increment photothermal conversion for certain multicore Au@mSiO₂ particles, in particular for those that exhibit a morphology compatible with hot spots. However, the frequency at which particles with this morphology appear in a sample is very low, and so no relevant change is expected for a photothermal experiment as the one done in Figure 6. As regards multicore NPs with more separated cores, they do not exhibit a representative variation in temperature at the surface of the shell with respect to 1 core Au@mSiO₂ NPs.

SUMMARY AND CONCLUSIONS

The mechanism behind one-pot syntheses for mesoporous silica nanoparticles containing one or more spherical gold cores was elucidated through multi-technique monitoring of the process. The process begins with a well-defined precursor state, composed of nonhydrolyzed TEOS, gold complexes, and clusters bound to the micelles. Any perturbation of this precursor state causes the reaction to follow a different path.

This is a key concept that limits the ways in which the final product can be modified using conventional strategies for $m\text{SiO}_2$ to fine-tune diameters and pore dimensions.

Upon the addition of a base, enough alcoholate is produced for Au(III) reduction and NP growth so that AuNPs develop in less than 6 s (*Stage I*) while the hydrolysis–condensation of TEOS leading to the mesoporous layer is detectable after 60 s (*Stage II*). During *Stage II*, the $m\text{SiO}_2$ shell continues to grow until *Stage III*, where particles consolidate invariantly in diameter and AuNPs become more spherical and mono-disperse.

This mechanism is described from the individual trajectories of $\text{Au(III)} \rightarrow \text{AuNP}$ and $\text{TEOS} \rightarrow m\text{SiO}_2$ which cross at some point in the reaction space. The crossover point would define the conditions under which the core–shell particles are built up. This scenario allows us to draw criteria for modifying the size and shape of the core, as well as the thickness of the shell.

Using an optimized combination of surfactants, base, and alcohols, we synthesize well-disperse particles with size below 100 nm in a robust and reproducible manner. The shell thickness can be fine-tuned, allowing for total particle size between 50 and 100 nm using the combination Brij C10:CTAB:TEOA (1:5:4) by controlling the timing of TEOA addition after triggering the reaction.

Attempts to widen pores with surfactants or cosolvents forming bigger micelles, as usually reported in templated sol–gel methods, alter the precursor state leading to NPs with large gold cores or gold located at the surface of silica particles. These examples highlight how such parameters can affect the chemistry of the gold species. Nevertheless, the core–shell $\text{Au}@m\text{SiO}_2$ particles with a thin mesoporous silica shell ($D_{\text{part}} = 50$ nm) can act as seeds to promote the formation of an additional layer of silica with large pores (8–13 nm) able to carry large biomolecules, which will be reported in a subsequent article.⁶²

Photothermal heating experiments demonstrated that optimized $\text{Au}@m\text{SiO}_2$ particles show a temperature increase under irradiation with light coincident with LSPR with an efficiency of ca 70% and a macroscopic temperature elevation of 18 °C under 0.8 W. However, simulations show that such nanoparticles exhibit a temperature increase of ca 0.35 °C at their surface under isothermal conditions for a continuous irradiance of 1 nW/nm², i.e., 1000 W mm^{−2}. This elevated irradiance necessary to heat demonstrates that the observed temperature increase of the suspension under excitation mostly results from the collective heating by a large number of nanoparticles ($\sim 10^{11}$ NP/mL), each of them providing a tiny heating effect. Even when two or three cores are present within such nanoparticles, simulations show that the temperature is only marginally increased. These results are therefore in contradiction with those that stipulate that such NPs can individually elevate their temperature by 40 °C at less than 1 W/mm², within a medium at constant temperature.^{31,70}

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsanm.4c07241>.

Data on reported one-pot methods and detailed information on experimental parameter influences, reaction monitoring processes, and additional characterizations (PDF)

Triggering the one-pot reaction (MP4)

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Author Contributions

The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript. S.A.B. and X.C. proposed the concepts and supervised the overall project. S.A.B., X.C., Y.R., and A.M.O. performed the syntheses, sample preparation, data characterization, and interpretation. M.L.M.R. and G.P.O. performed the photothermal response simulations. P.T. performed nitrogen sorption data acquisition, analysis, and interpretation. S.K. and M.C.M. perform microscopy characterizations, data analysis, and interpretation. All authors discussed the results and participated in writing the final version of the manuscript.

Funding

This work is supported by the French National Research Agency in the framework of the “Investissements d’avenir” program (ANR-15-IDEX-02), by Universidad de Buenos Aires, Argentina (UBACyT 20020190100299BA and 200202200115BA), by CONICET (PIP 112202101 00205), and by ANPCyT PICT 2021 I-A-0045 (without disbursements over 2024), by Universidad Nacional del Nordeste (PI-23F001-SGCyT-UNNE)

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

S.A.B. and M.L.M.R. thank the CNRS and the University Grenoble Alpes for invited professor grants. A.M.-O. thanks CONICET and University Grenoble Alpes for a PhD fellowship. M.L.M.R. and G.P.O. thank Dr. Luis W. Mochán for his useful comment on Photonic package use. Corinne Felix and Veronique Boutou (Inst Néel) are gratefully acknowledged for setting up the photothermal experiment. S.A.B. and A.M.-O. thank Bruno Aramburu (INQUIMAE) for enriching discussions on coordination compound spectroscopy. The authors are indebted to the Nanoandes Network and CNRS (IRN NanoAndes) for their continuous cooperative support.

ABBREVIATIONS

Au@mSiO₂, gold-mesoporous silica core–shell nanoparticles; NPs, nanoparticles; NVs, nanovehicles; mSiO₂, mesoporous silica; MNPs, metallic nanoparticles; LSPR, localized surface plasmon resonance; EM, electromagnetic; AuNPs, gold nanoparticles; LMCT, ligand to metal charge transfer

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